Alaa Badawii

Backend Software Engineer | Python Backend & AI-Powered Systems | FastAPI • Flask • SQL

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SUMMARY

Backend Software Engineer specializing in Python, API development, and database-driven applications. Experienced in building RESTful APIs, designing relational databases, implementing authentication and authorization, and managing transactions, concurrency, and data integrity. Strong focus on software reliability, automated testing, and backend architecture, with additional experience building AI-powered applications and integrating RAG and agentic workflows.

TECHNICAL SKILLS

Programming & Backend
Python · SQL · Flask · FastAPI · RESTful APIs · SQLAlchemy · Pydantic · OOP · JWT · RBAC · Async Programming · API Integration.

Databases & Data
MySQL · SQLite · Redis · PostgreSQL Fundamentals · MongoDB · Database Design · Transactions · Data Integrity · Concurrency · Idempotency.

Testing & Reliability
Unit Testing · Integration Testing · API Testing · Dependency Injection · Debugging · Logging · Failure Handling · System Evaluation · Retrieval Evaluation · Groundedness Evaluation · Citation Validation

Cloud, DevOps & Development Tools
Docker · Docker Compose · Linux · Bash · Git · GitHub · AWS EC2 · Postman.

AI & LLM Systems
RAG · LLM Applications · AI Agents · Agentic Workflows · LangChain · LangGraph · CrewAI · Tool Calling · Hybrid Retrieval · Cross-Encoder Reranking · Vector Databases · Chroma · LLM Evaluation · Citation Validation.

PROJECTS

Quizey — AI-First Assessment & Training Platform (In Progress)
https://github.com/AlaaBadawii/quizey-platform

- AI-first assessment and training platform designed to identify learners' strengths and gaps and support personalized, measurable improvement.
- Engineered a modular Python API with RBAC, JWT authentication, copy-on-write versioning, immutable exams, and configurable workflows.
- Built concurrency-safe grading and one-time submission using database-backed idempotency, atomic transactions, and frozen attempt papers.
- Built a comprehensive automated test suite with 1,100+ tests covering authentication, authorization, concurrency, data integrity, and DB migrations.

Team Lead — Grounded RAG over NICE NG217
https://github.com/AlaaBadawii/grounded-rag-ng217

- Built an end-to-end RAG system using hybrid retrieval, cross-encoder reranking, LLM generation, and deterministic citation validation.
- Built the FastAPI backend and Dockerized the application with Docker Compose, exposing the RAG pipeline through REST APIs.
- Evaluated retrieval and generation quality, achieving **85% P@1**, **87.9% MRR**, and **100% candidate recall**, with **75% answer correctness**, **100% grounded responses**, and **130/130 citations traceable** to retrieved evidence.

LinkedIn Personal Branding Agent — AI Content & Publishing Automation
https://github.com/AlaaBadawii/PersonalBrandingAgent

- Built a persistent, scheduled AI workflow for grounded LinkedIn content generation and publishing, combining RAG, LLM reasoning, and validation.
- Built a Chroma-based knowledge base and SQLite-backed state tracking for content intent, publication history, and duplicate prevention.
- Implemented publishing safeguards, validation, error handling, and notifications to prevent duplicate or unintended posts.

EXPERIENCE

Team Lead — Clinical RAG Hackathon Project | AI Hackathon
Banha, Egypt

Creativa AI Hackathon
Aug 2026 - Aug 2026

Led a 5-person team during a 40+ hour hackathon, coordinating backend, data, and frontend development of a clinical RAG system for NICE NG217.

- Coordinated technical delivery and retrieval improvements, contributing to an 85% Precision@1 result.

Backend Engineering Trainee
Remote

alx
Aug 2023 - Sep 2024

Completed ALX's Software Engineering programme, progressing through Foundations and Backend specialization.

- Developed backend applications using Python, Flask, SQLAlchemy, MySQL, REST APIs, authentication, caching, and automated testing.
- Applied software engineering practices in database design, Git, Linux, debugging, and application architecture through project-based learning.

EDUCATION

B.S. in Management Information Systems | Modern Academy
Oct 2019 - Jul 2023

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

IBM RAG and Agentic AI Professional Certificate (In Progress)
IBM | Coursera • 7 of 10 courses completed · Expected Nov 2026

Agentic AI with LangChain and LangGraph
IBM | Coursera • Oct *2026*

Practical DevOps Bootcamp for All Specialization
Packt | Coursera • Sep 2026

Introduction to FastAPI and Backend Development Fundamentals
Packt | Coursera • Aug 2026

System Design and Architecture
Manara • Mar 2026

ALX Africa — Certified Backend Software Engineer
ALX Africa • Aug 2023 – Sep 2024

LANGUAGES

Arabic (Native)

English (Professional Working Proficiency)